

SOLIDWORKS API (.NET VBA C#) — Reference **(43/337): ICoincidentMateFeatureData-IColorTable**

SOLIDWORKS API Documentation (converted from CHM)

ICoincidentMateFeatureData (Type)

ICoincidentMateFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICoincidentMateFeatureData Interface

Allows access to a coincident mate feature.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

NET Syntax

Visual Basic (Declaration)

```
Public Interface ICoincidentMateFeatureData
```

Visual Basic (Usage)



```
Dim instance As ICoincidentMateFeatureData
```

C#

```
public interface ICoincidentMateFeatureData
```

C++/CLI

```
public interface class ICoincidentMateFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [CoincidentMateFeatureData](#).

Example

[Create Standard Mates \(VBA\)](#)
[Create Standard Mates \(C#\)](#)

Remarks

A coincident mate:

- Positions the selected faces, edges, and planes so they share the same infinite plane.
- Positions two vertices so they touch.

[IMateFeatureData](#) is the parent of this mate interface.

To create a coincident mate:

- Follow general instructions in the [IAssemblyDoc::CreateMate](#) Remarks.
- Specify [ICoincidentMateFeatureData::EntitiesToMate](#) or use [IModelDocExtension::SelectByID2](#) to pre-select the entities to mate using Mark = 1.
- Specify other properties of the CoincidentMateFeatureData object.

To edit a coincident mate:

- Access its feature and call [IFeature::GetDefinition](#) to get the MateFeatureData object.
- Cast the MateFeatureData object to a CoincidentMateFeatureData object.
- You cannot change the entities to mate by pre-selection. You must use ICoincidentMateFeatureData::EntitiesToMate to modify them.
- Modify other properties of the CoincidentMateFeatureData object.
- Call [IFeature::ModifyDefinition](#).

To delete this coincident mate, use [IModelDocExtension::DeleteSelection2](#).

Access Diagram

[CoincidentMateFeatureData](#)

See Also

[ICoincidentMateFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

ICoincidentMateFeatureData Interface Members

[See Also](#) [Properties](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICoincidentMateFeatureData Interface

The following tables list the members exposed by [ICoincidentMateFeatureData](#).

Public Properties

Name	Description
 EntitiesToMate	Gets or sets the entities to mate in this coincident mate.
 MateAlignment	Gets or sets the mate alignment of this coincident mate.
 PickPoints	Gets or sets the pick points for the entities to mate in this coincident mate.

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See Also

[ICoincidentMateFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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ICoincidentMateFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICoincidentMateFeatureData Interface

For a list of all members of this type, see [ICoincidentMateFeatureData members](#).

Public Properties

Name	Description
 EntitiesToMate	Gets or sets the entities to mate in this coincident mate.
 MateAlignment	Gets or sets the mate alignment of this coincident mate.
 PickPoints	Gets or sets the pick points for the entities to mate in this coincident mate.

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See Also

[ICoincidentMateFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

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EntitiesToMate Property (ICoincidentMateFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICoincidentMateFeatureData Interface](#) : EntitiesToMate Property (ICoincidentMateFeatureData)

Gets or sets the entities to mate in this coincident mate.

.NET Syntax

Visual Basic (Declaration)

```
Property EntitiesToMate As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICoincidentMateFeatureData
Dim value As System.Object

instance.EntitiesToMate = value
value = instance.EntitiesToMate
```

C#

```
System.object EntitiesToMate {get; set;}
```

C++/CLI

```
property System.Object^ EntitiesToMate {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of entities to mate (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [CoincidentMateFeatureData::EntitiesToMate](#).

Example

See the [ICoincidentMateFeatureData](#) example.

Remarks

Instead of specifying this property, you can use [IModelDocExtension::SelectById2](#) to pre-select the entities to mate using Mark = 1. You can pre-select mate entities during mate creation, but not during mate editing.

Populate the array of this property by using either [IModelDocExtension::SelectById2](#) or [IModelDocExtension::SelectByRay](#) and [ISelectionMgr::GetSelectedObject6](#).

Create coincident mates with the following mate entity combinations:

Mate entity/Mate entity	IEdge (arc, circular edge)	ISurface, IFace2 (cone)	IFeature (coordinate system)	ICurve (arc, spline, helix)	ISurface, IFace2 (cylinder)	IFace2 (extrusion face, draft is not allowed)	IEdge, IRefAxis, ICenterLine, ISketchLine (line)	IEntity (coordinate system origin)	IRefPlane IRefPoint, IVertex, ISketchPoint	ISurface, IFace2 (sphere)	ISurface
IEdge (arc, circular edge)	•	•			•				•	•	
ISurface, IFace2 (cone)	•	• Both cones must be of the same half angle								•	
IFeature (coordinate system)			•					•			
ICurve (arc, spline, helix)										•	
ISurface, IFace2 (cylinder)	•						•				
IFace2 (extrusion face, draft is not allowed)										•	
IEdge, IRefAxis, ICenterLine, ISketchLine (line)					•		•		•	•	
IEntity (coordinate system origin)			•					•			
IRefPlane	•								•	•	
IRefPoint, IVertex, ISketchPoint	•	•		•	•	•	•		•	•	•
ISurface, IFace2 (sphere)										•	
ISurface										•	

After specifying the entities to mate, specify their [ICoincidentMateFeatureData::PickPoints](#) to fully define the position of the mate.

See Also

[ICoincidentMateFeatureData Interface](#)
[ICoincidentMateFeatureData Members](#)

Availability

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MateAlignment Property (ICoincidentMateFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICoincidentMateFeatureData Interface](#) : MateAlignment Property (ICoincidentMateFeatureData)

Gets or sets the mate alignment of this coincident mate.

NET Syntax

Visual Basic (Declaration)

Property MateAlignment As System.Integer

Visual Basic (Usage)



```
Dim instance As ICoincidentMateFeatureData
Dim value As System.Integer

instance.MateAlignment = value

value = instance.MateAlignment
```

C#

```
System.int MateAlignment {get; set;}
```

C++/CLI

```
property System.int MateAlignment {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Mate alignment as defined in [swMateAlign_e](#)

Visual Basic for Applications (VBA) Syntax

See [CoincidentMateFeatureData::MateAlignment](#).

Example

See the [ICoincidentMateFeatureData](#) example.

See Also

[ICoincidentMateFeatureData Interface](#)
[ICoincidentMateFeatureData Members](#)

Availability

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PickPoints Property (ICoincidentMateFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICoincidentMateFeatureData Interface](#) : PickPoints Property (ICoincidentMateFeatureData)

Gets or sets the pick points for the entities to mate in this coincident mate.

NET Syntax

Visual Basic (Declaration)

Property PickPoints As System.Object

Visual Basic (Usage)

```
Dim instance As ICoincidentMateFeatureData  
Dim value As System.Object  
  
instance.PickPoints = value  
  
value = instance.PickPoints
```

C#

```
System.object PickPoints {get; set;}
```

C++/CLI

```
property System.Object^ PickPoints {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of x, y, z-coordinates, one set for each mate entity in [ICoincidentMateFeatureData::EntitiesToMate](#)

Visual Basic for Applications (VBA) Syntax

See [CoincidentMateFeatureData::PickPoints](#).

Remarks

Pick points define the position of the entities to mate. If no pick points are specified, then entities are mated using default pick points.

After you create this mate using the pick points for a given selection of `ICoincidentMateFeatureData::EntitiesToMate`, you cannot edit the pick points. If you want to use new pick points, then you must create an entirely new coincident mate with mate entities and pick points.

If you pre-select the mate entities by coordinates (e.g., [IModelDocExtension::SelectByID2](#)), then you do not need to specify pick points.

See Also

[ICoincidentMateFeatureData Interface](#)
[ICoincidentMateFeatureData Members](#)

Availability

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ICollision (Type)

ICollision Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICollision Interface

Allows access to collision data.
NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

NET Syntax

Visual Basic (Declaration)

Public Interface ICollision

Visual Basic (Usage)



Dim instance As [ICollision](#)

C#

public interface ICollision

C++/CLI

public interface class ICollision

Visual Basic for Applications (VBA) Syntax

See [Collision](#).

Example

See the [ICollisionDetectionManager](#) example.

Remarks

See the ICollisionDetectionManager remarks.

Accessors

[ICollisionDetectionManager::GetCollisions](#)

Access Diagram

[Collision](#)

See Also

[ICollision Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ICollisionDetectionGroup Interface](#)
[Interference Interface](#)

ICollision Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICollision Interface

For a list of all members of this type, see [ICollision members](#).

Public Methods

Name	Description
 GetComponents	Gets the components involved in this collision.
 GetTransforms	Gets the current transforms of the components involved in this collision.
 IsPenetrating	Gets whether the components involved in this collision penetrate one another.

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See Also

[ICollision Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ICollisionDetectionGroup Interface](#)
[Interference Interface](#)

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ICollision Interface Members

[See Also](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : ICollision Interface

The following tables list the members exposed by [ICollision](#).

Public Methods

Name	Description
 GetComponents	Gets the components involved in this collision.
 GetTransforms	Gets the current transforms of the components involved in this collision.
 IsPenetrating	Gets whether the components involved in this collision penetrate one another.

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See Also

[ICollision Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ICollisionDetectionGroup Interface](#)
[Interference Interface](#)

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GetComponents Method (ICollision)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICollision Interface](#) : GetComponents Method (ICollision)

Gets the components involved in this collision.

.NET Syntax

Visual Basic (Declaration)

Function GetComponents() As System.Object

Visual Basic (Usage)



```
Dim instance As ICollision  
Dim value As System.Object  
  
value = instance.GetComponents()
```

C#

System.object GetComponents()

C++/CLI

System.Object^ GetComponents();

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [IComponent2](#)

Visual Basic for Applications (VBA) Syntax

See [Collision::GetComponents](#).

See Also

[ICollision Interface](#)
[ICollision Members](#)

Availability

GetTransforms Method (ICollision)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [ICollision Interface](#) : GetTransforms Method (ICollision)

Gets the current transforms of the components involved in this collision.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTransforms() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICollision
Dim value As System.Object

value = instance.GetTransforms()
```

C#

```
System.object GetTransforms()
```

C++/CLI

```
System.Object^ GetTransforms();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [IMathTransform](#)

Visual Basic for Applications (VBA) Syntax

See [Collision::GetTransforms](#).

See Also

[ICollision Interface](#)
[ICollision Members](#)

Availability

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IsPenetrating Method (ICollision)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [ICollision Interface](#) : IsPenetrating Method (ICollision)

Gets whether the components involved in this collision penetrate one another.

.NET Syntax

Visual Basic (Declaration)

```
Function IsPenetrating() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICollision
Dim value As System.Boolean

value = instance.IsPenetrating()
```

C#

```
System.bool IsPenetrating()
```

C++/CLI

```
System.bool IsPenetrating();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if components penetrate one another, false if they are only in contact

Visual Basic for Applications (VBA) Syntax

See [Collision::IsPenetrating](#).

Example

See the [ICollisionDetectionManager](#) example.

See Also

[ICollision Interface](#)
[ICollision Members](#)

Availability

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ICollisionDetectionGroup (Type)

ICollisionDetectionGroup Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICollisionDetectionGroup Interface

Allows access to collision detection groups.
NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

Visual Basic (Declaration)

Public Interface ICollisionDetectionGroup

Visual Basic (Usage)



Dim instance As [ICollisionDetectionGroup](#)

C#

public interface ICollisionDetectionGroup

C++/CLI

public interface class ICollisionDetectionGroup

Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionGroup](#).

Example

See the [ICollisionDetectionManager](#) example.

Remarks

See the ICollisionDetectionManager remarks.

Accessors

[ICollisionDetectionManager::CreateGroup](#)

[ICollisionDetectionManager::GetGroup](#)

Access Diagram

[CollisionDetectionGroup](#)

See Also

[ICollisionDetectionGroup Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ICollision Interface](#)

ICollisionDetectionGroup Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICollisionDetectionGroup Interface

The following tables list the members exposed by [ICollisionDetectionGroup](#).

Public Methods

Name	Description
 ApplyTransforms	Applies the specified transforms to the components in this collision detection group.
 GetComponentCount	Gets the number of components in this collision detection group.
 GetComponents	Gets the components in this collision detection group.
GetTransforms	Gets the current transforms of the components in this collision detection group.

		
	RemoveAllTransforms	Removes all collision detection transforms from all components in this collision detection group and restores the original transforms.
	SetComponents	Sets the specified components in this collision detection group.

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See Also

[ICollisionDetectionGroup Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ICollision Interface](#)

SOLIDWORKS API Help

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ICollisionDetectionGroup Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICollisionDetectionGroup Interface

For a list of all members of this type, see [ICollisionDetectionGroup members](#).

Public Methods

Name	Description
 ApplyTransforms	Applies the specified transforms to the components in this collision detection group.
 GetComponentCount	Gets the number of components in this collision detection group.
 GetComponents	Gets the components in this collision detection group.
 GetTransforms	Gets the current transforms of the components in this collision detection group.
 RemoveAllTransforms	Removes all collision detection transforms from all components in this collision detection group and restores the original transforms.
 SetComponents	Sets the specified components in this collision detection group.

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See Also

[ICollisionDetectionGroup Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ICollision Interface](#)

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[Send comments on this topic.](#)

ApplyTransforms Method (ICollisionDetectionGroup)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICollisionDetectionGroup Interface](#) : ApplyTransforms Method (ICollisionDetectionGroup)

Applies the specified transforms to the components in this collision detection group.

.NET Syntax

Visual Basic (Declaration)

```
Function ApplyTransforms( _  
    ByVal ComponentTransforms As System.Object _  
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As ICollisionDetectionGroup  
Dim ComponentTransforms As System.Object  
Dim value As System.Integer  
  
value = instance.ApplyTransforms(ComponentTransforms)
```

C#

```
System.int ApplyTransforms(  
    System.object ComponentTransforms  
)
```

C++/CLI

```
System.int ApplyTransforms(  
& System.Object^ ComponentTransforms  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ComponentTransforms
Array of [IMathTransform](#)

Return Value

Result code as defined in [swCollisionGroupApplyTransformErrors_e](#)

Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionGroup::ApplyTransforms](#).

Example

See the [ICollisionDetectionManager](#) example.

See Also

[ICollisionDetectionGroup Interface](#)
[ICollisionDetectionGroup Members](#)

Availability

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SOLIDWORKS API Help

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GetComponentCount Method (ICollisionDetectionGroup)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICollisionDetectionGroup Interface](#) : GetComponentCount Method (ICollisionDetectionGroup)

Gets the number of components in this collision detection group.

.NET Syntax

Visual Basic (Declaration)

```
Function GetComponentCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICollisionDetectionGroup  
Dim value As System.Integer  
  
value = instance.GetComponentCount()
```

C#

```
System.int GetComponentCount()
```

C++/CLI

```
System.int GetComponentCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of components

Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionGroup::GetComponentCount](#).

See Also

[ICollisionDetectionGroup Interface](#)
[ICollisionDetectionGroup Members](#)

Availability

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GetComponents Method (ICollisionDetectionGroup)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICollisionDetectionGroup Interface](#) : GetComponent Method (ICollisionDetectionGroup)

Gets the components in this collision detection group.

NET Syntax

Visual Basic (Declaration)

```
Function GetComponent() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICollisionDetectionGroup
Dim value As System.Object

value = instance.GetComponent()
```

C#

```
System.object GetComponent()
```

C++/CLI

```
System.Object^ GetComponent();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [IComponent2](#)

Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionGroup.GetComponent](#).

Example

See the [ICollisionDetectionManager](#) example.

See Also

[ICollisionDetectionGroup Interface](#)
[ICollisionDetectionGroup Members](#)

Availability

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SOLIDWORKS API Help

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GetTransforms Method (ICollisionDetectionGroup)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICollisionDetectionGroup Interface](#) : GetTransforms Method (ICollisionDetectionGroup)

Gets the current transforms of the components in this collision detection group.

NET Syntax

Visual Basic (Declaration)

```
Function GetTransforms() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICollisionDetectionGroup
Dim value As System.Object

value = instance.GetTransforms()
```

C#

```
System.object GetTransforms()
```

C++/CLI

```
System.Object^ GetTransforms();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [IMathTransform](#)

 Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionGroup::GetTransforms](#).

 See Also

[ICollisionDetectionGroup Interface](#)
[ICollisionDetectionGroup Members](#)

 Availability

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RemoveAllTransforms Method (ICollisionDetectionGroup)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICollisionDetectionGroup Interface](#) : RemoveAllTransforms Method (ICollisionDetectionGroup)

Removes all collision detection transforms from all components in this collision detection group and restores the original transforms.

 .NET Syntax

Visual Basic (Declaration)

```
Sub RemoveAllTransforms()
```

Visual Basic (Usage)



```
Dim instance As ICollisionDetectionGroup  
instance.RemoveAllTransforms()
```

C#

```
void RemoveAllTransforms()
```

C++/CLI

```
void RemoveAllTransforms();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

 Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionGroup::RemoveAllTransforms](#).

 See Also

[ICollisionDetectionGroup Interface](#)
[ICollisionDetectionGroup Members](#)

 Availability

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SOLIDWORKS API Help

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SetComponents Method (ICollisionDetectionGroup)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICollisionDetectionGroup Interface](#) : SetComponents Method (ICollisionDetectionGroup)

Sets the specified components in this collision detection group.

 .NET Syntax

Visual Basic (Declaration)

```
Function SetComponents( _  
    ByVal Components As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICollisionDetectionGroup  
Dim Components As System.Object  
Dim value As System.Integer  
  
value = instance.SetComponents(Components)
```

C#

```
System.int SetComponents(  
    System.object Components  
)
```

C++/CLI

```
System.int SetComponents(  
& System.Object^ Components  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Components
Array of [IComponent2](#)

Return Value

Return code as defined in [swCollisionGroupSetComponentsErrors_e](#)

Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionGroup::SetComponents](#).

Example

See the [ICollisionDetectionManager](#) example.

See Also

[ICollisionDetectionGroup Interface](#)
[ICollisionDetectionGroup Members](#)

Availability

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ICollisionDetectionManager (Type)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICollisionDetectionManager Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICollisionDetectionManager Interface

Allows access to the collision detection manager.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

Visual Basic .NET Syntax

Visual Basic (Declaration)

Public Interface ICollisionDetectionManager

Visual Basic (Usage)



Dim instance As [ICollisionDetectionManager](#)

C#

public interface ICollisionDetectionManager

C++/CLI

public interface class ICollisionDetectionManager

Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionManager](#).

Example

'VBA

'This example demonstrates how to set up one collision detection frame.

- '1. Open *public_documents\samples\tutorial\api\Spindleassem.sldasm*.
- '2. Run the macro below.

```
Dim swApp As SldWorks.SldWorks
Dim Part As SldWorks.ModelDoc2
Dim swDoc As SldWorks.AssemblyDoc
Dim cdm As SldWorks.CollisionDetectionManager
Dim cdg As SldWorks.CollisionDetectionGroup
Dim cdg2 As SldWorks.CollisionDetectionGroup
Dim cdg3 As SldWorks.CollisionDetectionGroup
Dim aCollision As SldWorks.Collision
Dim TransformData(15) As Double
Dim TransformDataVariant As Variant
Dim TransformData1(15) As Double
Dim TransformDataVariant1 As Variant
Dim comp(0) As SldWorks.Component2
Dim comp1(0) As SldWorks.Component2
Dim comp2(0) As SldWorks.Component2
Dim transform(0) As SldWorks.MathTransform
Dim transform1(0) As SldWorks.MathTransform
Dim var1 As Variant
Dim boolstatus As Boolean
Dim longstatus As Long
Dim numCollisions As Long
Dim i As Long
Dim j As Long
Dim comps As Variant
Option Explicit
Sub main()

    Set swApp = Application.SldWorks
    Set Part = swApp.ActiveDoc
    Set swDoc = Part
    Set cdm = swApp.GetCollisionDetectionManager
    longstatus = cdm.SetAssembly(swDoc)

    cdm.GraphicsRedrawEnabled = True

    Set cdg = cdm.CreateGroup
    Set cdg2 = cdm.CreateGroup
```

```

Set cdg3 = cdm.CreateGroup

'Holder
boolstatus = Part.Extension.SelectByID2("Holder-1@Spindleassem", "COMPONENT", 0, 0, 0, True, 0, Nothing, 0)

TransformData(0) = 0.561729092855705
TransformData(1) = -0.827321235216108
TransformData(2) = 0
TransformData(3) = 0.827321235216108
TransformData(4) = 0.561729092855705
TransformData(5) = 0
TransformData(6) = 0
TransformData(7) = 0
TransformData(8) = 1#
TransformData(9) = 2.02097529022064E-02
TransformData(10) = 0.025899850979138
TransformData(11) = 0.1
TransformData(12) = 1
TransformData(13) = 0
TransformData(14) = 0
TransformData(15) = 0

TransformDataVariant = TransformData

'Cutting tool
boolstatus = Part.Extension.SelectByID2("CuttingTool-1@Spindleassem", "COMPONENT", 0, 0, 0, True, 0, Nothing, 0)

TransformData1(0) = 1
TransformData1(1) = 0
TransformData1(2) = 0
TransformData1(3) = 0
TransformData1(4) = 1
TransformData1(5) = 0
TransformData1(6) = 0
TransformData1(7) = 0
TransformData1(8) = 1
TransformData1(9) = -0.01
TransformData1(10) = -0.03
TransformData1(11) = 0.1
TransformData1(12) = 1
TransformData1(13) = 0
TransformData1(14) = 0
TransformData1(15) = 0

TransformDataVariant1 = TransformData1

'Work piece
boolstatus = Part.Extension.SelectByID2("workpiece-1@Spindleassem", "COMPONENT", 0, 0, 0, True, 0, Nothing, 0)

Set comp(0) = Part.SelectionManager.GetSelectedObjectsComponent4(1, -1)
Set comp1(0) = Part.SelectionManager.GetSelectedObjectsComponent4(2, -1)
Set comp2(0) = Part.SelectionManager.GetSelectedObjectsComponent4(3, -1)

Dim swMathUtil As Object
Set swMathUtil = swApp.GetMathUtility()

Set transform(0) = swMathUtil.CreateTransform((TransformDataVariant))
Set transform1(0) = swMathUtil.CreateTransform((TransformDataVariant1))

longstatus = cdg.SetComponents(comp) 'Holder
longstatus = cdg2.SetComponents(comp1) 'Cutting tool
longstatus = cdg3.SetComponents(comp2) 'Work piece

'Move the cutting tool and work piece

longstatus = cdg.ApplyTransforms(transform)
longstatus = cdg2.ApplyTransforms(transform1)

'Detect collisions

longstatus = cdm.IsCollisionPresent(True) ' Treat coincidence as a collision
numCollisions = cdm.GetCollisions(True, var1)

If numCollisions > 0 Then
    For i = 0 To (numCollisions - 1)
        Debug.Print "Collision " & (i + 1)
        Set aCollision = var1(i)
        Debug.Print " Is penetrating? " & aCollision.IsPenetrating
        comps = aCollision.GetComponents()
        For j = 0 To UBound(comps)
            Debug.Print " " & comps(j).Name
        Next j
    Next i
Else
    Debug.Print "No collisions present"
End If

End Sub

```

Example

[Manage Collision Detection \(C#\)](#)

Remarks

The purpose of this interface, [ICollisionDetectionGroup](#), and [ICollision](#) is to provide dynamic, high-efficiency collision detection among groups of components. Collision detection is performed exclusively between collision groups: neither within collision groups nor outside of them.

Please use [InterferenceDetectionMgr](#) if you need to perform static interference detection between components.

Collision detection may be thought of as a series of collision detection frames. Each collision detection frame begins by setting a target assembly and two or more collision groups of components that are transformed to move to precise locations in model space. A collision detection frame may or may not end with a collision.

NOTE: If a collision is detected, an [ICollision](#) object is created. Be aware that [ICollision](#) objects are volatile and should not be cached for later review. [ICollision](#) objects are invalidated when a new collision detection frame begins or whenever you:

- call [ICollisionDetectionManager::IsCollisionPresent](#).
- call [ICollisionDetectionManager::GetCollisions](#).
- apply new transforms to collision group components.
- create new or modify existing collision groups.
- set a new target assembly.

Because you cannot save [ICollision](#) objects for later review, you should record the collision detection frame setup and any resulting collision components and transforms before the next collision detection frame begins.

The workflow to detect collisions:

1. Call [ICollisionDetectionManager::SetAssembly](#) to set the target assembly for collision detection.
2. Call [ICollisionDetectionManager::CreateGroup](#) to create two collision groups.
3. Call [ICollisionDetectionGroup::SetComponents](#) to populate each collision group.
4. Use [ICollisionDetectionGroup::ApplyTransforms](#) to transform components into working positions.
5. Call [ICollisionDetectionManager::IsCollisionPresent](#) and/or [ICollisionDetectionManager::GetCollisions](#) to detect collisions.
6. Inspect any resulting [ICollision](#) objects. (See the NOTE above.)
7. Repeat steps 4-6 if you want to apply new transforms for the next collision detection frame.
8. Repeat steps 2-6 if you want to change collision groups.
9. Repeat steps 1-6 if you want to change the target assembly.

Accessors

[ISldWorks::GetCollisionDetectionManager](#)

Access Diagram

[CollisionDetectionManager](#)

See Also

[ICollisionDetectionManager Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICollisionDetectionManager Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) ; [ICollisionDetectionManager](#) Interface

The following tables list the members exposed by [ICollisionDetectionManager](#).

Public Properties

Name	Description
 GraphicsRedrawEnabled	Gets or sets whether to display components in their transformed positions.

[Top](#)

Public Methods

Name	Description
 CreateGroup	Creates a collision detection group of non-colliding components.
 GetAssembly	Gets the active assembly for this collision detection manager.
 GetCollisions	Gets the collisions detected.
 GetGroup	Gets the specified collision detection group.
 GetGroupCount	Gets the number of collision detection groups currently defined.
 IsCollisionPresent	Performs collision detection analysis between all groups of components.
 RemoveGroup	Removes the specified collision detection group from collision detection.
 SetAssembly	Sets the active assembly for this collision detection manager.

[Top](#)

See Also

[ICollisionDetectionManager Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICollisionDetectionManager Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICollisionDetectionManager Interface

For a list of all members of this type, see [ICollisionDetectionManager members](#).

Public Methods

Name	Description
 CreateGroup	Creates a collision detection group of non-colliding components.
 GetAssembly	Gets the active assembly for this collision detection manager.
 GetCollisions	Gets the collisions detected.
 GetGroup	Gets the specified collision detection group.
 GetGroupCount	Gets the number of collision detection groups currently defined.
 IsCollisionPresent	Performs collision detection analysis between all groups of components.
 RemoveGroup	Removes the specified collision detection group from collision detection.
 SetAssembly	Sets the active assembly for this collision detection manager.

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See Also

[ICollisionDetectionManager Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICollisionDetectionManager Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ICollisionDetectionManager Interface

For a list of all members of this type, see [ICollisionDetectionManager members](#).

Public Properties

Name	Description
 GraphicsRedrawEnabled	Gets or sets whether to display components in their transformed positions.

[Top](#)

See Also

[ICollisionDetectionManager Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateGroup Method (ICollisionDetectionManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICollisionDetectionManager Interface](#) : CreateGroup Method (ICollisionDetectionManager)

Creates a collision detection group of non-colliding components.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateGroup() As System.Object
```

Visual Basic (Usage)

```
Dim instance As ICollisionDetectionManager  
Dim value As System.Object  
  
value = instance.CreateGroup()
```



C#

```
System.Object CreateGroup()
```

C++/CLI

```
System.Object^ CreateGroup();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[ICollisionDetectionGroup](#)

Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionManager::CreateGroup](#).

Example

See the [ICollisionDetectionManager](#) example.

See Also

[ICollisionDetectionManager Interface](#)
[ICollisionDetectionManager Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetAssembly Method (ICollisionDetectionManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICollisionDetectionManager Interface](#) : GetAssembly Method (ICollisionDetectionManager)

Gets the active assembly for this collision detection manager.

.NET Syntax

Visual Basic (Declaration)

```
Function GetAssembly() As System.Object
```

Visual Basic (Usage)

```
Dim instance As ICollisionDetectionManager  
Dim value As System.Object  
  
value = instance.GetAssembly()
```

C#

```
System.Object GetAssembly()
```

C++/CLI

```
System.Object^ GetAssembly();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IAssemblyDoc](#)

Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionManager::GetAssembly](#).

See Also

[ICollisionDetectionManager Interface](#)
[ICollisionDetectionManager Members](#)
[ICollisionDetectionManager::SetAssembly Method\(\)](#)

Availability

SOLIDWORKS 2019 FCS Revision Number 27.0

[Send comments on this topic.](#)

SOLIDWORKS API Help

GetCollisions Method (ICollisionDetectionManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICollisionDetectionManager Interface](#) : GetCollisions Method (ICollisionDetectionManager)

Gets the collisions detected.

.NET Syntax

Visual Basic (Declaration)

```
Function GetCollisions( _  
    ByVal TreatContactAsCollision As System.Boolean, _  
    ByRef Collisions As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICollisionDetectionManager  
Dim TreatContactAsCollision As System.Boolean  
Dim Collisions As System.Object  
Dim value As System.Integer  
  
value = instance.GetCollisions(TreatContactAsCollision, Collisions)
```

C#

```
System.int GetCollisions(  
    System.bool TreatContactAsCollision,  
    out System.object Collisions  
)
```

C++/CLI

```
System.int GetCollisions(  
& System.bool TreatContactAsCollision,  
& [Out] System.Object^ Collisions  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TreatContactAsCollision
True to treat touching faces/edges/vertices as colliding, false to require solid bodies to overlap in a finite volume
Collisions
Array of [Collisions](#)

Return Value

Number of collisions found

Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionManager::GetCollisions](#).

Example

See the [ICollisionDetectionManager](#) example.

Remarks

Use [ICollisionDetectionManager::IsCollisionPresent](#) to determine whether to run this method.

This method can take significantly longer to execute than [ICollisionDetectionManager::IsCollisionPresent](#), because this method might perform the collision detection calculation repeatedly.

See Also

[ICollisionDetectionManager Interface](#)
[ICollisionDetectionManager Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

[Send comments on this topic.](#)

SOLIDWORKS API Help

GetGroup Method (ICollisionDetectionManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICollisionDetectionManager Interface](#) : GetGroup Method (ICollisionDetectionManager)

Gets the specified collision detection group.

.NET Syntax

Visual Basic (Declaration)

```
Function GetGroup( _  
    ByVal GroupIndex As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ICollisionDetectionManager  
Dim GroupIndex As System.Integer  
Dim value As System.Object  
  
value = instance.GetGroup(GroupIndex)
```

C#

```
System.object GetGroup(  
    System.int GroupIndex  
)
```

C++/CLI

```
System.Object^ GetGroup(  
& System.int GroupIndex  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

GroupIndex
Zero-based index of collision detection group

Return Value

[ICollisionDetectionGroup](#)

Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionManager::GetGroup](#).

See Also

[ICollisionDetectionManager Interface](#)
[ICollisionDetectionManager Members](#)

Availability

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SOLIDWORKS API Help

GetGroupCount Method (ICollisionDetectionManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICollisionDetectionManager Interface](#) : GetGroupCount Method (ICollisionDetectionManager)

Gets the number of collision detection groups currently defined.

.NET Syntax

Visual Basic (Declaration)

```
Function GetGroupCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICollisionDetectionManager  
Dim value As System.Integer  
  
value = instance.GetGroupCount()
```

C#

```
System.int GetGroupCount()
```

C++/CLI

```
System.int GetGroupCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of collision detection groups

Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionManager::GetGroupCount](#).

See Also

[ICollisionDetectionManager Interface](#)
[ICollisionDetectionManager Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GraphicsRedrawEnabled Property (ICollisionDetectionManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICollisionDetectionManager Interface](#) : GraphicsRedrawEnabled Property (ICollisionDetectionManager)

Gets or sets whether to display components in their transformed positions.

.NET Syntax

Visual Basic (Declaration)

```
Property GraphicsRedrawEnabled As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICollisionDetectionManager  
Dim value As System.Boolean  
  
instance.GraphicsRedrawEnabled = value  
value = instance.GraphicsRedrawEnabled
```

C#

```
System.bool GraphicsRedrawEnabled {get; set;}
```

C++/CLI

```
property System.bool GraphicsRedrawEnabled {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to display components in their transformed positions, false to display components in their last known positions

Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionManager::GraphicsRedrawEnabled](#).

Remarks

If this property is set to false, the graphics display update is drastically reduced. The client application may force the assembly to be redrawn by calling [IModelDoc2::GraphicsRedraw2](#).

Set this property to false for maximum performance.

See Also

[ICollisionDetectionManager Interface](#)
[ICollisionDetectionManager Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsCollisionPresent Method (ICollisionDetectionManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICollisionDetectionManager Interface](#) : IsCollisionPresent Method (ICollisionDetectionManager)

Performs collision detection analysis between all groups of components.

.NET Syntax

Visual Basic (Declaration)

```
Function IsCollisionPresent( _  
    ByVal TreatContactAsCollision As System.Boolean _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICollisionDetectionManager  
Dim TreatContactAsCollision As System.Boolean  
Dim value As System.Integer  
  
value = instance.IsCollisionPresent(TreatContactAsCollision)
```

C#

```
System.int IsCollisionPresent(  
    System.bool TreatContactAsCollision  
)
```

C++/CLI

```
System.int IsCollisionPresent(  
& System.bool TreatContactAsCollision  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TreatContactAsCollision

True to treat touching faces/edges/vertices as colliding, false to require solid bodies to overlap in a finite volume

Return Value

Return code as defined in [swCollisionDetectionResults_e](#)

Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionManager::IsCollisionPresent](#).

Example

See the [ICollisionDetectionManager](#) example.

Remarks

This method takes less time to run than [ICollisionDetectionManager::GetCollisions](#).

See Also

[ICollisionDetectionManager Interface](#)
[ICollisionDetectionManager Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RemoveGroup Method (ICollisionDetectionManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICollisionDetectionManager Interface](#) : RemoveGroup Method (ICollisionDetectionManager)

Removes the specified collision detection group from collision detection.

.NET Syntax

Visual Basic (Declaration)

```
Function RemoveGroup( _  
    ByVal GroupIndex As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ICollisionDetectionManager  
Dim GroupIndex As System.Integer  
Dim value As System.Boolean  
  
value = instance.RemoveGroup(GroupIndex)
```

C#

```
System.bool RemoveGroup(  
    System.int GroupIndex  
)
```

C++/CLI

```
System.bool RemoveGroup(  
&    System.int GroupIndex  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

GroupIndex
Zero-based index of the collision detection group to remove from collision detection

Return Value

True if collision detection group successfully removed, false if not

Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionManager.RemoveGroup](#).

See Also

[ICollisionDetectionManager Interface](#)
[ICollisionDetectionManager Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetAssembly Method (ICollisionDetectionManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ICollisionDetectionManager Interface](#) : SetAssembly Method (ICollisionDetectionManager)

Sets the active assembly for this collision detection manager.

.NET Syntax

Visual Basic (Declaration)

```
Function SetAssembly( _  
    ByVal OwnerAssem As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ICollisionDetectionManager  
Dim OwnerAssem As System.Object  
Dim value As System.Integer  
  
value = instance.SetAssembly(OwnerAssem)
```

C#

```
System.int SetAssembly(  
    System.object OwnerAssem  
)
```

C++/CLI

```
System.int SetAssembly(  
& System.Object^ OwnerAssem  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[OwnerAssem](#)
[IAsemblyDoc](#)

Return Value

Error code as defined in [swCollisionManagerSetAssemblyErrors_e](#)

Visual Basic for Applications (VBA) Syntax

See [CollisionDetectionManager::SetAssembly](#).

Example

See the [ICollisionDetectionManager](#) examples.

See Also

[ICollisionDetectionManager Interface](#)
[ICollisionDetectionManager Members](#)
[ICollisionDetectionManager::GetAssembly Method \(\)](#)

Availability

SOLIDWORKS 2019 FCS Revision Number 27.0

IColorTable (Type)

IColorTable Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IColorTable Interface

Allows access to the color definitions used in SOLIDWORKS.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IColorTable
```

Visual Basic (Usage)



```
Dim instance As IColorTable
```

C#

```
public interface IColorTable
```

C++/CLI

```
public interface class IColorTable
```

Visual Basic for Applications (VBA) Syntax

See [ColorTable](#).

Example

[Get COLORREF Values of Standard User-interface Elements \(C#\)](#)
[Get COLORREF Values of Standard User-interface Elements \(VB.NET\)](#)
[Get COLORREF Values of Standard User-interface Elements \(VBA\)](#)

Remarks

This API uses COLORREF values, which are equivalent to the data type long. See MSDN for details on how to create and interpret COLORREF values.

Accessors

[ISldWorks::GetColorTable](#) and [ISldWorks::IGetColorTable](#)

Access Diagram

[ColorTable](#)

See Also

[IColorTable Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IColorTable Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IColorTable Interface

For a list of all members of this type, see [IColorTable members](#).

Public Methods

Name	Description
GetBasicColorCount	Gets the number of basic colors defined in the color table.
GetBasicColors	Gets the basic colors in the color table.
GetColorRefAtIndex	Gets the COLORREF at the specified index position of the color table.
GetCount	Gets the total number of colors in the color table, including standard and custom colors.

 GetCustomColorCount	Gets the number of custom colors in the color table.
 GetCustomColors	Gets the number of custom colors in the color table.
 GetNameAtIndex	Gets the name of the color at the specified index position in the color table.
 GetStandardCount	Gets the number of basic or standard colors defined in the color table.
 IGetBasicColors	Gets the basic colors in the color table.
 IGetCustomColors	Gets the number of custom colors in the color table.
 SetColorRefAtIndex	Sets the specified color value within the color table.
 SetCustomColor	Sets a custom color.

[Top](#)

See Also

[IColorTable Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IColorTable Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IColorTable Interface

The following tables list the members exposed by [IColorTable](#).

Public Methods

Name	Description
 GetBasicColorCount	Gets the number of basic colors defined in the color table.
 GetBasicColors	Gets the basic colors in the color table.
 GetColorRefAtIndex	Gets the COLORREF at the specified index position of the color table.
 GetCount	Gets the total number of colors in the color table, including standard and custom colors.
 GetCustomColorCount	Gets the number of custom colors in the color table.
 GetCustomColors	Gets the number of custom colors in the color table.
 GetNameAtIndex	Gets the name of the color at the specified index position in the color table.
 GetStandardCount	Gets the number of basic or standard colors defined in the color table.
 IGetBasicColors	Gets the basic colors in the color table.
 IGetCustomColors	Gets the number of custom colors in the color table.
 SetColorRefAtIndex	Sets the specified color value within the color table.
 SetCustomColor	Sets a custom color.

[Top](#)

See Also

[IColorTable Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetBasicColorCount Method (IColorTable)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IColorTable Interface](#) : GetBasicColorCount Method (IColorTable)

Gets the number of basic colors defined in the color table.

NET Syntax

Visual Basic (Declaration)

```
Function GetBasicColorCount() As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IColorTable  
Dim value As System.Integer
```



```
value = instance.GetBasicColorCount()
```

C#

```
System.int GetBasicColorCount()
```

C++/CLI

```
System.int GetBasicColorCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of basic colors

 Visual Basic for Applications (VBA) Syntax

See [ColorTable::GetBasicColorCount](#).

 Remarks

Call this method before calling [IColorTable::IGetBasicColors](#) to determine the size of the array for that method.

 See Also

[IColorTable Interface](#)
[IColorTable Members](#)
[IColorTable::IGetBasicColors Method](#)
[IColorTable::GetBasicColors Method](#)
[IColorTable::GetCount Method](#)
[IColorTable::GetCustomColorCount Method](#)
[IColorTable::GetStandardCount Method](#)

 Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetBasicColors Method (IColorTable)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IColorTable Interface](#) : GetBasicColors Method (IColorTable)

Gets the basic colors in the color table.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetBasicColors() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IColorTable  
Dim value As System.Object  
  
value = instance.GetBasicColors()
```

C#

```
System.object GetBasicColors()
```

C++/CLI

```
System.Object^ GetBasicColors();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array containing the basic colors

 Visual Basic for Applications (VBA) Syntax

See [ColorTable::GetBasicColors](#).

See Also

[IColorTable Interface](#)
[IColorTable Members](#)
[IColorTable::GetBasicColorCount Method](#)
[IColorTable::IGetBasicColors Method](#)
[IColorTable::GetCustomColors Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetColorRefAtIndex Method (IColorTable)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IColorTable Interface](#) : GetColorRefAtIndex Method (IColorTable)

Gets the COLORREF at the specified index position of the color table.

.NET Syntax

Visual Basic (Declaration)

```
Function GetColorRefAtIndex( _  
    ByVal Index As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IColorTable  
Dim Index As System.Integer  
Dim value As System.Integer  
  
value = instance.GetColorRefAtIndex(Index)
```

C#

```
System.int GetColorRefAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.int GetColorRefAtIndex(  
    & System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index position of the desired color

Return Value

COLORREF at the specified index position

Visual Basic for Applications (VBA) Syntax

See [ColorTable::GetColorRefAtIndex](#).

Example

[Get COLORREF Values of Standard User-interface Elements \(C#\)](#)
[Get COLORREF Values of Standard User-interface Elements \(VB.NET\)](#)
[Get COLORREF Values of Standard User-interface Elements \(VBA\)](#)

See Also

[IColorTable Interface](#)
[IColorTable Members](#)
[IColorTable::SetColorRefAtIndex Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetCount Method (IColorTable)

[See Also](#)



Gets the total number of colors in the color table, including standard and custom colors.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IColorTable
Dim value As System.Integer

value = instance.GetCount()
```

C#

```
System.int GetCount()
```

C++/CLI

```
System.int GetCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of colors defined in the color table

 Visual Basic for Applications (VBA) Syntax

See [ColorTable::GetCount](#).

 See Also

[IColorTable Interface](#)
[IColorTable Members](#)
[IColorTable::GetBasicColorCount Method](#)
[IColorTable::GetCustomColorCount Method](#)
[IColorTable::GetStandardCount Method](#)

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GetCustomColorCount Method (IColorTable)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > IColorTable Interface : GetCustomColorCount Method (IColorTable)

Gets the number of custom colors in the color table.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetCustomColorCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IColorTable
Dim value As System.Integer

value = instance.GetCustomColorCount()
```

C#

```
System.int GetCustomColorCount()
```

C++/CLI

```
System.int GetCustomColorCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of custom colors

 Visual Basic for Applications (VBA) Syntax

See [ColorTable::GetCustomColorCount](#).

Remarks

Call this method before calling [IColorTable::IGetCustomColors](#) to determine the size of the array for that method.

See Also

[IColorTable Interface](#)
[IColorTable Members](#)
[IColorTable::GetBasicColorCount Method](#)
[IColorTable::GetCount Method](#)
[IColorTable::GetCustomColors Method](#)
[IColorTable::GetStandardCount Method](#)

Availability

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GetCustomColors Method (IColorTable)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IColorTable Interface](#) : GetCustomColors Method (IColorTable)

Gets the number of custom colors in the color table.

NET Syntax

Visual Basic (Declaration)

```
Function GetCustomColors() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IColorTable
Dim value As System.Object

value = instance.GetCustomColors()
```

C#

```
System.Object GetCustomColors()
```

C++/CLI

```
System.Object^ GetCustomColors();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array containing the custom colors

Visual Basic for Applications (VBA) Syntax

See [ColorTable::GetCustomColors](#).

See Also

[IColorTable Interface](#)
[IColorTable Members](#)
[IColorTable::GetCustomColorCount Method](#)
[IColorTable::IGetCustomColors Method](#)
[IColorTable::SetCustomColor Method](#)
[IColorTable::GetBasicColors Method](#)
[IColorTable::GetCount Method](#)

Availability

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GetNameAtIndex Method (IColorTable)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IColorTable Interface](#) : GetNameAtIndex Method (IColorTable)

Gets the name of the color at the specified index position in the color table.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetNameAtIndex(  
    ByVal Index As System.Integer  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IColorTable  
Dim Index As System.Integer  
Dim value As System.String  
  
value = instance.GetNameAtIndex(Index)
```

C#

```
System.string GetNameAtIndex(  
    System.int Index  
)
```

C++/CLI

```
System.String^ GetNameAtIndex(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index position of the desired color

Return Value

Name of the color at the specified index

Visual Basic for Applications (VBA) Syntax

See [ColorTable::GetNameAtIndex](#).

Example

[Get COLORREF Values of Standard User-interface Elements \(C#\)](#)
[Get COLORREF Values of Standard User-interface Elements \(VB.NET\)](#)
[Get COLORREF Values of Standard User-interface Elements \(VBA\)](#)

See Also

[IColorTable Interface](#)
[IColorTable Members](#)
[IColorTable::SetColorRefAtIndex Method](#)

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GetStandardCount Method (IColorTable)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IColorTable Interface](#) : GetStandardCount Method (IColorTable)

Gets the number of basic or standard colors defined in the color table.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetStandardCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IColorTable  
Dim value As System.Integer  
  
value = instance.GetStandardCount()
```

C#

```
System.int GetStandardCount()
```

C++/CLI

```
System.int GetStandardCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of basic or standard colors defined in the color table

Visual Basic for Applications (VBA) Syntax

See [ColorTable::GetStandardCount](#).

Example

[Get COLORREF Values of Standard User-interface Elements \(C#\)](#)
[Get COLORREF Values of Standard User-interface Elements \(VB.NET\)](#)
[Get COLORREF Values of Standard User-interface Elements \(VBA\)](#)

See Also

[IColorTable Interface](#)
[IColorTable Members](#)
[IColorTable::GetBasicColorCount Method](#)
[IColorTable::GetCustomColorCount Method](#)
[IColorTable::GetCount Method](#)

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IGetBasicColors Method (IColorTable)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IColorTable Interface](#) : IGetBasicColors Method (IColorTable)

Gets the basic colors in the color table.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetBasicColors( _  
    ByVal ColorCount As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IColorTable  
Dim ColorCount As System.Integer  
Dim value As System.Integer  
  
value = instance.IGetBasicColors(ColorCount)
```

C#

```
System.int IGetBasicColors(  
    System.int ColorCount  
)
```

C++/CLI

```
System.int IGetBasicColors(  
    & System.int ColorCount  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ColorCount
Number of basic colors

Return Value

- in-process, unmanaged C++: Pointer to an array of basic colors of size ColorCount
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IColorTable::GetBasicColorCount](#) to determine the size of the array.

See Also

[IColorTable Interface](#)
[IColorTable Members](#)
[IColorTable::GetBasicColors Method](#)
[IColorTable::IGetCustomColors Method](#)

Availability

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IGetCustomColors Method (IColorTable)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IColorTable Interface](#) : IGetCustomColors Method (IColorTable)

Gets the number of custom colors in the color table.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetCustomColors( _  
    ByVal ColorCount As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IColorTable  
Dim ColorCount As System.Integer  
Dim value As System.Integer  
  
value = instance.IGetCustomColors(ColorCount)
```

C#

```
System.int IGetCustomColors(  
    System.int ColorCount  
)
```

C++/CLI

```
System.int IGetCustomColors(  
&    System.int ColorCount  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ColorCount

Number of custom colors

Return Value

- in-process, unmanaged C++: Pointer to an array of custom colors of size ColorCount
- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IColorTable::GetCustomColorCount](#) to determine the size of the array.

See Also

[IColorTable Interface](#)
[IColorTable Members](#)
[IColorTable::GetCustomColors Method](#)
[IColorTable::SetCustomColor Method](#)

Availability

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SetColorRefAtIndex Method (IColorTable)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IColorTable Interface](#) : SetColorRefAtIndex Method (IColorTable)

Sets the specified color value within the color table.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetColorRefAtIndex( _  
    ByVal Index As System.Integer, _  
    ByVal ColorRef As System.Integer, _  
    ByVal ApplyTo As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IColorTable  
Dim Index As System.Integer  
Dim ColorRef As System.Integer  
Dim ApplyTo As System.Integer  
  
instance.SetColorRefAtIndex(Index, ColorRef, ApplyTo)
```

C#

```
void SetColorRefAtIndex(  
    System.int Index,  
    System.int ColorRef,  
    System.int ApplyTo  
)
```

C++/CLI

```
void SetColorRefAtIndex(  
& System.int Index,  
& System.int ColorRef,  
& System.int ApplyTo  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Index value within the color table you want to modify

ColorRef
COLORREF value

ApplyTo
Not used; specify 0

Visual Basic for Applications (VBA) Syntax

See [ColorTable::SetColorRefAtIndex](#).

See Also

[IColorTable Interface](#)
[IColorTable Members](#)
[IColorTable::SetColorRefAtIndex Method](#)

Availability

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SetCustomColor Method (IColorTable)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IColorTable Interface](#) : SetCustomColor Method (IColorTable)

Sets a custom color.

.NET Syntax

Visual Basic (Declaration)

```
Function SetCustomColor( _  
    ByVal Index As System.Integer, _  
    ByVal ColorRef As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IColorTable  
Dim Index As System.Integer  
Dim ColorRef As System.Integer  
Dim value As System.Boolean  
  
value = instance.SetCustomColor(Index, ColorRef)
```

C#

```
System.bool SetCustomColor(  
    System.int Index,  
    System.int ColorRef  
)
```

C++/CLI

```
System.bool SetCustomColor(  
& System.int Index,  
& System.int ColorRef  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the custom color to set

ColorRef

COLORREF value

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ColorTable::SetCustomColor](#).

See Also

[IColorTable Interface](#)
[IColorTable Members](#)
[IColorTable::GetCustomColors Method](#)
[IColorTable::IGetCustomColors Method](#)
[IColorTable::GetCustomColorCount Method](#)

Availability

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